Please check the examination details below before entering your candidate information

Candidate surname	Other names

Group

Candidate Number

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Expected Paper 2

November 2026

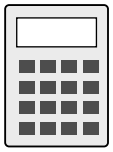
Time: **2 hours**

Paper reference **ELT/U2X2**

Mathematics A

MODULAR · UNIT 2

Higher Tier · Expected Exam



Calculator allowed

You must have: Ruler graduated in centimetres and millimetres, protractor, pair of compasses, pen, HB pencil, eraser, calculator.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, group and candidate number.
- Answer **all** questions.
- Without sufficient working, correct answers may be awarded no marks.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- **Calculators may be used.**
- You must **NOT** write anything on the formulae page.

Information

- The total mark for this paper is 100.
- There are **26** questions in this paper.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Questions 19 to 26 are the most demanding. Aim to reach them with at least 45 minutes left.
- Check your answers if you have time at the end.

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Dr Eslam Ahmed
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ELITE IGCSE ACADEMY

ELT U2X2 2026

International GCSE Mathematics

Formulae sheet – Higher Tier

Arithmetic series

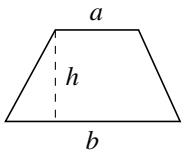
Sum to n terms, $S_n = \frac{n}{2}[2a + (n-1)d]$

The quadratic equation

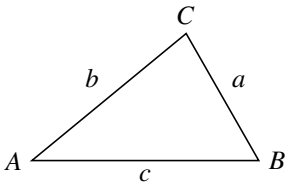
The solutions of $ax^2 + bx + c = 0$ where $a \neq 0$ are given by:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Area of trapezium $= \frac{1}{2}(a+b)h$



Trigonometry



In any triangle ABC

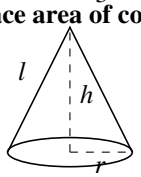
Sine Rule $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

Cosine Rule $a^2 = b^2 + c^2 - 2bc\cos A$

Area of triangle $= \frac{1}{2}ab\sin C$

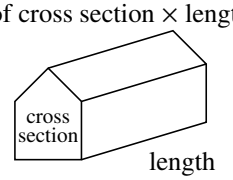
Volume of cone $= \frac{1}{3}\pi r^2 h$

Curved surface area of cone $= \pi r l$



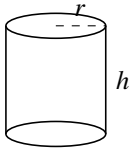
Volume of prism

= area of cross section × length



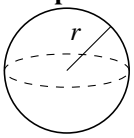
Volume of cylinder $= \pi r^2 h$

Curved surface area of cylinder $= 2\pi r h$



Volume of sphere $= \frac{4}{3}\pi r^3$

Surface area of sphere $= 4\pi r^2$



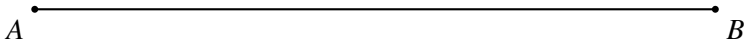
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Answer ALL TWENTY SIX questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

1 Use ruler and compasses only to construct an angle of 30° at the point A on the line AB.
You must show all your construction lines.



(Total for Question 1 is 2 marks)

2 The table gives information about the times, t minutes, that 30 boys spent reading on Monday.

Time (t minutes)	Frequency
$0 < t \leqslant 10$	4
$10 < t \leqslant 20$	11
$20 < t \leqslant 30$	9
$30 < t \leqslant 40$	6

The mean time that 20 girls spent reading on Monday was 14.5 minutes.

Work out an estimate for the mean time spent reading on Monday by all 50 of these children.

. . . . . minutes

(4)

(Total for Question 2 is 4 marks)

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3 A virus is 1.2×10^{-7} metres long.
Viruses of this type are placed end to end to make a straight line 3 cm long.

Work out the number of viruses in the line.
Give your answer in standard form.

.....

(3)

(Total for Question 3 is 3 marks)

---

4 The highest common factor (HCF) of two numbers is 12
The lowest common multiple (LCM) of the two numbers is 180
Both numbers are greater than 12

Find the two numbers.

..... and .....

(3)

(Total for Question 4 is 3 marks)

---

5 The straight line with equation $ax + by = 12$ passes through the points $(-2, 9)$ and $(6, -3)$

Find the value of a and the value of b.
Show clear algebraic working.

$a =$ .....

$b =$ .....

(3)

(Total for Question 5 is 3 marks)

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6 Green paint is made by mixing blue paint and yellow paint in the ratio

blue : yellow = 2 : 3

Sana has 1.2 litres of blue paint and 1.5 litres of yellow paint.

Work out the greatest amount of green paint that Sana can make.

. . . . . litres

(3)

(Total for Question 6 is 3 marks)

---

7 Leo buys a car for \$18 000

The value of the car depreciates by 15% each year.

Work out the least number of whole years it takes for the value of the car to fall below half of the price that Leo paid.

. . . . . years

(3)

(Total for Question 7 is 3 marks)

---

8 P is a regular polygon with n sides.

Q is a regular polygon with $2n$ sides.

The size of each interior angle of Q is 15° more than the size of each interior angle of P.

Work out the value of n.

n = ...

(3)

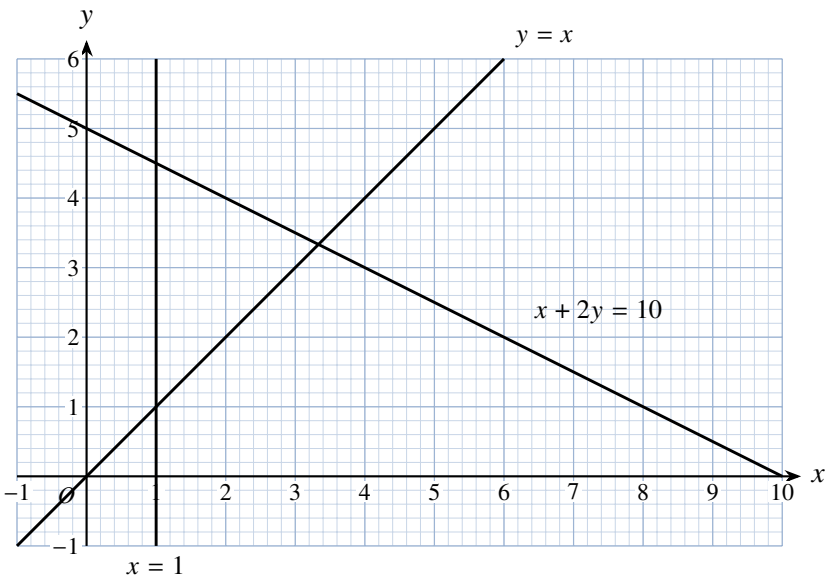
(Total for Question 8 is 3 marks)

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9 The lines with equations $x = 1$, $y = x$ and $x + 2y = 10$ are drawn on the grid.



(a) Show, by shading on the grid, the region that satisfies all three of the inequalities

$$x \geqslant 1 \qquad y \geqslant x \qquad x + 2y \leqslant 10$$

Label the region R.

(2)

(b) The point (x, y) lies in the region R. x and y are both integers.
Work out the greatest possible value of $2x + y$

.....

(2)

(Total for Question 9 is 4 marks)

---

10 The price of a bike was increased by x%.
The new price of the bike was then decreased by x%.
After these two changes, the price of the bike was 4% less than its original price.
Work out the value of x.

$x =$ .....

(3)

(Total for Question 10 is 3 marks)

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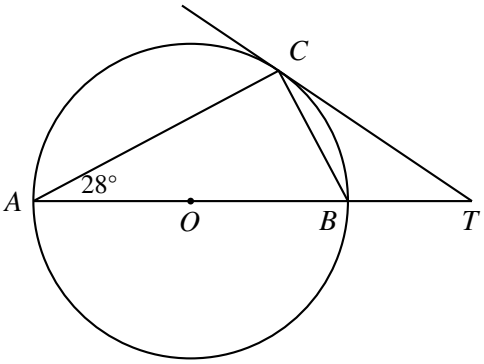
11 A, B and C are points on a circle, centre O.
AB is a diameter of the circle.

The tangent to the circle at C meets AB extended at the point T.

Angle $CAB = 28°$

Work out the size of angle ATC.
Give a reason for each stage of your working.

Diagram **NOT** accurately drawn



. . . . . °

(4)

(Total for Question 11 is 4 marks)

---

12 The time, T seconds, for one complete swing of a pendulum is directly proportional to the square root of L, the length of the pendulum in centimetres.

A pendulum of length 25 cm takes 1 second for one complete swing.

(a) Find a formula for T in terms of L.

$T =$ . . . . . .

(2)

(b) Complete the table.

Length L (cm)	25	64	
Time T (seconds)	1		1.8

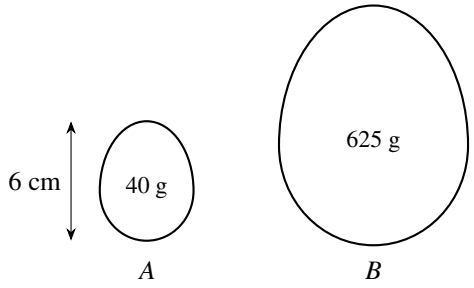
(2)

(Total for Question 12 is 4 marks)

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13 Two solid chocolate eggs, A and B, are mathematically similar.
They are made from the same chocolate.
Egg A has a mass of 40 g and a height of 6 cm.
Egg B has a mass of 625 g.

Diagram **NOT** accurately drawn



(a) Work out the height of egg B.

..... cm
(2)

(b) Egg B is wrapped in 200 cm^2 of foil.
Work out the area of foil needed to wrap egg A.

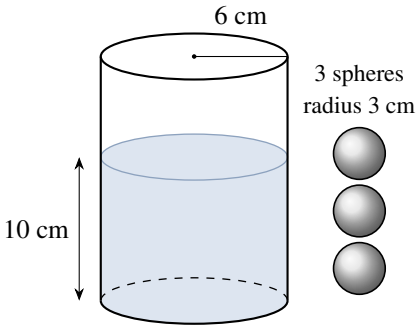
..... cm^2
(2)

(Total for Question 13 is 4 marks)

---

14 A cylinder has radius 6 cm.
It contains water to a depth of 10 cm.
Three identical solid metal spheres, each of radius 3 cm, are put into the water.
The spheres sink and are completely covered by the water. No water spills out.
Work out the new depth of the water.

Diagram **NOT** accurately drawn



..... cm
(3)

(Total for Question 14 is 3 marks)

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15 The table gives information about the times, t minutes, that 80 people took to complete a task.

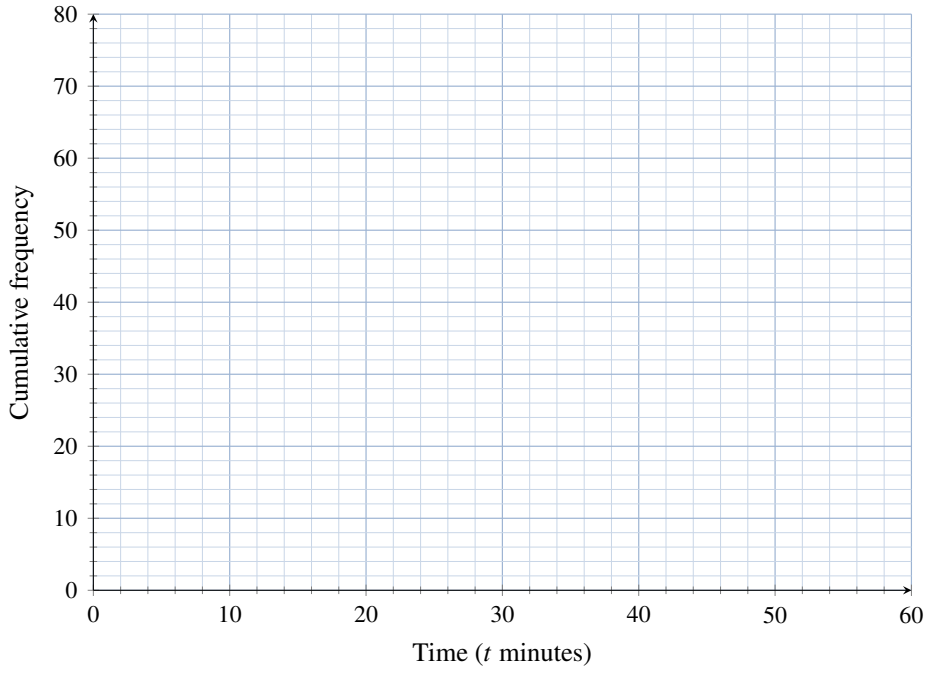
Time (t minutes)	Frequency
$0 < t \leqslant 10$	6
$10 < t \leqslant 20$	14
$20 < t \leqslant 30$	25
$30 < t \leqslant 40$	20
$40 < t \leqslant 50$	11
$50 < t \leqslant 60$	4

Time (t minutes)	Cumulative frequency
$0 < t \leqslant 10$	
$0 < t \leqslant 20$	
$0 < t \leqslant 30$	
$0 < t \leqslant 40$	
$0 < t \leqslant 50$	
$0 < t \leqslant 60$	

(a) Complete the cumulative frequency table.

(1)

(b) On the grid, draw a cumulative frequency graph for your table.



(2)

(c) The 12 people who took the longest time are asked to do the task again.
Use your graph to find an estimate for the shortest time taken by one of these 12 people.

. . . . . minutes

(2)

(Total for Question 15 is 5 marks)

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16 Make a the subject of the formula $t = \dfrac{ab}{a+b}$

$a =$ .....

(4)

(Total for Question 16 is 4 marks)

---

17 The functions f and g are such that

$$f(x) = 2x + k \qquad g(x) = x^2 - 3$$

where k is a constant.

$fg(3) = 17$

(a) Find the value of k.

$k =$ .....

(2)

(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = \ldots$

$f^{-1}(x) =$ .....

(1)

(c) Solve $f^{-1}(x) = g(x)$
Show clear algebraic working.

.....

(2)

(Total for Question 17 is 5 marks)

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18 A ball is thrown upwards from the ground.
The height, h metres, of the ball above the ground t seconds after it is thrown is given by

$$h = 20t - 5t^2$$

Work out the length of time for which the ball is more than 15 metres above the ground.
Show clear algebraic working.

. . . . . seconds

(3)

(Total for Question 18 is 3 marks)

---

19 A particle P is moving along a straight line that passes through the fixed point O.
The displacement, s metres, of P from O at time t seconds is given by

$$s = t^3 - 9t^2 + 24t \qquad t \geqslant 0$$

(a) Find the two times at which P is instantaneously at rest.
Show clear algebraic working.

. . . . . .

(3)

(b) Work out the distance travelled by P between these two times.

. . . . . m

(2)

(Total for Question 19 is 5 marks)

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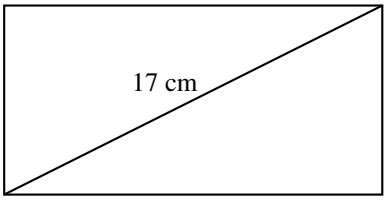
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20 A rectangle has a perimeter of 46 cm.
A diagonal of the rectangle is 17 cm long.

Work out the area of the rectangle.
Show clear algebraic working.

Diagram **NOT** accurately drawn



. . . . . cm^2

(5)

(Total for Question 20 is 5 marks)

---

21 The sum of the first n terms of an arithmetic series is given by

$$S_n = 2n^2 + 7n$$

(a) Find the first term and the common difference of the series.

first term . . . . . .

common difference . . . . . .

(2)

(b) Work out the sum of the 11th term to the 20th term of the series, inclusive.

. . . . . .

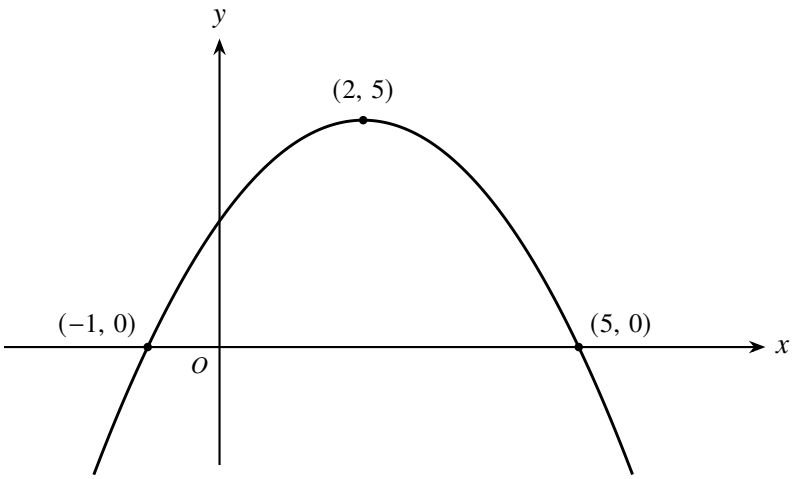
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(Total for Question 21 is 4 marks)

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22 The diagram shows a sketch of the curve with equation $y = f(x)$
The curve crosses the x-axis at the points $(-1, 0)$ and $(5, 0)$
The curve has a maximum point at $(2, 5)$



(a) Write down the coordinates of the maximum point of the curve with equation $y = 2f(x)$

(. . . . . , ...)

(1)

(b) The curve with equation $y = f(x + a)$ passes through the origin O.
Find the two possible values of a.

. . . . . .

(2)

(c) The maximum point of the curve with equation $y = f(kx)$ is $(0.5, 5)$
Find the value of k.

k = ...

(1)

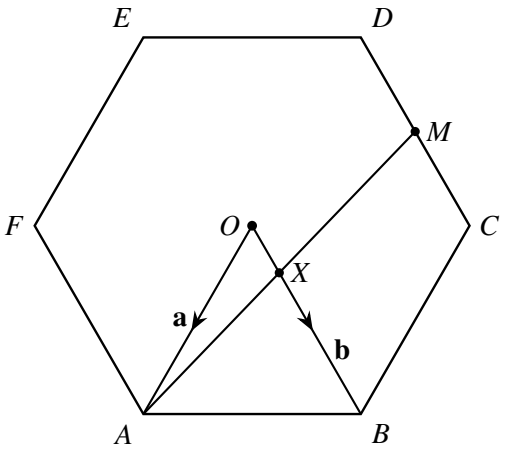
(Total for Question 22 is 4 marks)

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23 $ABCDEF$ is a regular hexagon with centre O.

$\overrightarrow{OA} = \mathbf{a}$ $\overrightarrow{OB} = \mathbf{b}$

Diagram **NOT** accurately drawn



(a) Find $\overrightarrow{OC}$ in terms of **a** and **b**.

. . . . . .

(1)

M is the midpoint of CD.
The line AM meets OB at the point X.

(b) Using a vector method, find the ratio $OX : XB$
Show your working clearly.

. . . . . : . . . . . .

(4)

(Total for Question 23 is 5 marks)

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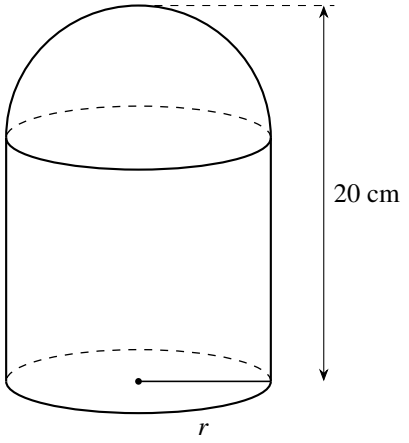
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24 A solid is made from a cylinder and a hemisphere.
The flat face of the hemisphere is joined to the top of the cylinder.
The cylinder and the hemisphere both have radius r cm.

The total height of the solid is 20 cm.
The total surface area of the solid is 225π cm^2

Work out the volume of the solid.
Give your answer correct to 3 significant figures.

Diagram **NOT** accurately drawn



. . . . . cm^3

(5)

(Total for Question 24 is 5 marks)

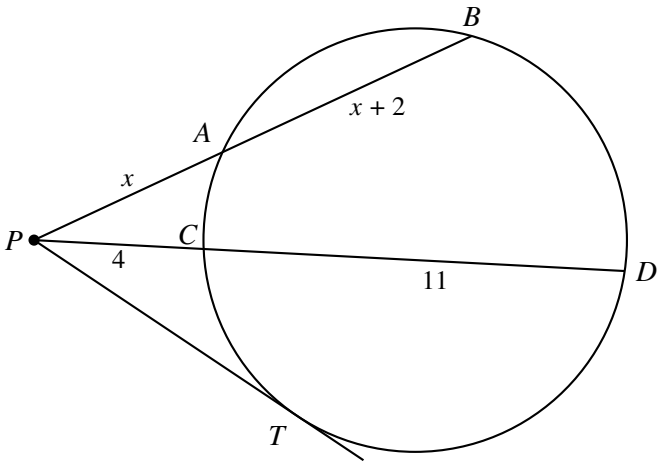
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25 A, B, C, D and T are points on a circle.
PAB and PCD are straight lines.
PT is the tangent to the circle at T.

Diagram **NOT** accurately drawn

$PA = x$ cm $\quad AB = (x + 2)$ cm

$PC = 4$ cm $\quad CD = 11$ cm



(a) Work out the value of x.
Show clear algebraic working.

$x =$ .....

(3)

(b) Work out the length of PT.
Give your answer correct to 3 significant figures.

..... cm

(1)

(Total for Question 25 is 4 marks)

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26 The curve C has equation

$$y = x^3 + ax^2 + bx + 1$$

where a and b are constants.

The point $(1, -4)$ is a turning point on C.

(a) Find the value of a and the value of b.
Show clear algebraic working.

$a =$ .....

$b =$ .....

(3)

(b) Find the coordinates of the other turning point on C.

(....., ...)

(2)

(Total for Question 26 is 5 marks)

TOTAL FOR PAPER IS 100 MARKS

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